

Volume 1 - AI Fundamentals & Architecture

Purpose

This volume covers the foundations required for enterprise AI engineering interviews.

Topics

AI, Machine Learning, Generative AI, LLMs, Tokens, Context Windows, Embeddings, RAG, Azure AI Search, Semantic Kernel, Prompt Engineering, HR Copilot Architecture.

Sample Questions

1. Explain the end-to-end architecture of your HR AI Copilot.

Answer: Describe Angular → ASP.NET Core → AI Orchestrator → Semantic Kernel → Plugins → SQL/Azure AI Search → Azure OpenAI.

2. Why use RAG instead of training an LLM?

Answer: RAG keeps knowledge current, avoids retraining, reduces hallucinations, and works with private enterprise data.

3. What are embeddings?

Answer: Numerical vector representations used for semantic similarity search.